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BRACKET FOR ADJUSTABLE-DEPTH MOUNTING OF ELECTRICAL BOXES

Abstract

A bracket for adjustable-depth mounting of an electrical box relative to a support structure can include a support body and a support arm. The support body can be configured to be secured to the support structure. A fastener can be retained on the support arm and can be adjusted relative to the support arm to move the electrical box to be secured at any of a plurality of depths relative to the support body.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/669,450, filed May 20, 2024, which is a continuation of U.S. patent application Ser. No. 18/457,902, filed Aug. 29, 2023, which is a continuation of U.S. patent application Ser. No. 16/952,197, filed November 19, 2020, which claims priority to U.S. Provisional Patent Applications Nos. 62/941,492 and 62/987,041, filed Nov. 27, 2019 and Mar. 9, 2020, respectively, all of which are incorporated herein by reference in their entireties.

BACKGROUND

[0002] In many applications, it may be useful to support electrical boxes at different depths relative to support structures, such as ceiling or wall structures of buildings. For example, in order to dispose electrical boxes of different depths to be flush with internal wall or ceiling surfaces, it may be useful to support the electrical boxes at different depths relative to a particular support structure.

SUMMARY

[0003] Some embodiments of the invention provide a bracket for adjustable-depth mounting of an electrical box relative to a support structure. The bracket can include a support body that is configured to be secured to the support structure, a support arm, and a fastener that is retained on the support arm and is configured to permit adjustment of the fastener relative to the support arm. The support arm can be configured to hold the fastener relative to the electrical box, so that the adjustment of the fastener relative to the support arm moves the electrical box to be secured at any of a plurality of depths relative to the support body.

[0004] Some embodiments of the invention provide a bracket for adjustable-depth mounting of an electrical box relative to a support structure. The bracket can include a support body and a support arm. The support body can be configured to be secured to the support structure and to surround exterior sides of the electrical box. The support arm can extend integrally from the support body and can include an extension portion and a support portion. The extension portion can be configured to extend from the support body along the exterior of a side of the electrical box. The support portion can extend at an angle from an end of the extension portion opposite the support body and can be configured to retain a threaded fastener to the exterior of a back side of the electrical box with the threaded fastener extending into engagement with a fastener opening on the electrical box so that rotational adjustment of the fastener relative to the support portion moves the electrical box to be secured at any of a plurality of depths relative to the support body.

[0005] Some embodiments of the invention provide a method of adjusting a mounting depth of an electrical box relative to a bracket. The electrical box can be secured to the bracket using a fastener secured to a support arm of the bracket. The fastener can be rotated relative to the support arm to move the electrical box to any of a plurality of depths relative to a support body of the bracket.

[0006] Some embodiments of the invention provide a bracket assembly for adjustable-depth mounting of an electrical box relative to a support structure. A support body can be configured to be secured to the support structure, the support body including a body opening that is fully enclosed by the support body and sized to slidably receive the electrical box. First and second support arms can extend integrally from respective edges of the support opening to respective free ends and can include respective bends configured to dispose the free end behind respective corners

of a rear wall of the electrical box, with a support portion proximate each of the free ends rotatably supporting a corresponding threaded fastener that is configured to threadedly engage with the electrical box to be rotatable from inside the electrical box to adjust a depth of the electrical box within the body opening. A plurality of attachment arms, including at least one of a spring arm or a bendable tab, can extend integrally from the support body and can be configured to collectively secure the support body to the support structure. A plurality of guide tabs can extend integrally from the support body into alignment with the body opening, the guide tabs being configured to biasingly engage side walls of the electrical box to guide movement of the electrical box within the opening.

[0007] Some embodiments of the invention provide a bracket assembly for adjustable-depth mounting of an electrical box relative to a support structure. A support body can be configured to be secured to the support structure and can include a body opening sized to slidably receive the electrical box. A first support arm and a second support arm can extend from the support body on opposing sides of the body opening, with the first support arm extending independently relative to the second support arm. Each of the first and second support arms can have a support portion configured to extend behind the electrical box when the electrical box is slidably received in the body opening, the end portion retaining a fastener that is configured to be adjusted from inside the electrical box, relative to the corresponding first or second support arm to move the electrical box to any of a plurality of depths relative to the support body.

[0008] Some embodiments of the invention provide a manufacturing blank for a bracket assembly. The manufacturing blank can include a single-piece planar body that includes a central body opening, first and second supports, and first and second sets of arms. The central body opening can be fully enclosed by material of the single-piece planar body. The first and second supports can extend into the central body opening from opposing edges of the central body opening. The first and second sets of arms can extend away from the central body opening, from each of two opposing sides of the single-piece planar body. Each of the first and second sets of arms can include a first attachment arm that extends farther away from the central body opening than second and third attachment arms of the first or second set of arms and that includes a free end with an extended tab opposite the central body opening

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The accompanying drawings, which are incorporated in and form a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of embodiments of the invention:

[0010] FIG. 1 is an isometric view of a ceiling structure and a bracket according to an embodiment of the invention, with the bracket supporting an electrical box relative to the ceiling structure;

[0011] FIG. 2 is an isometric sectional partial view of the ceiling structure and bracket of FIG. 1 taken along plane 2-2 of FIG. 1;

[0012] FIG. 3 is an isometric partial view of a ceiling structure and another bracket according to an embodiment of the invention, with the bracket supporting an electrical box relative to the ceiling structure;

[0013] FIGS. 4 and 5 are isometric partial views of a support structure and a bracket according to an embodiment of the invention, with the bracket supporting an electrical box relative to the support structure;

[0014] FIG. 6 is an isometric partial view of a ceiling structure and a bracket according to an embodiment of the invention, with the bracket supporting an electrical box relative to the ceiling structure;

[0015] FIG. 7 is a front plan view of another bracket according to an embodiment of the invention, with the bracket supporting an electrical box;

[0016] FIG. 8 is an isometric partial view of the bracket of FIG. 7 and a support structure, with the bracket supporting the electrical box relative to the support structure;

[0017] FIG. 9 is an isometric view of another bracket according to an embodiment of the invention, with the bracket supporting an electrical box;

[0018] FIG. 10 is a front elevation view of the bracket of FIG. 9;

[0019] FIG. 11 is a top plan view of the bracket of FIG. 9, installed on a support structure; and

[0020] FIG. 12 is a top plan view of a manufacturing blank for a bracket according to an embodiment of the invention.

DETAILED DESCRIPTION

[0021] Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings. Further, “connected” and “coupled” are not restricted to physical or mechanical connections or couplings.

[0022] Also as used herein, unless otherwise specified or limited, directional terms are presented only with regard to the particular embodiment and perspective described. For example, reference to features or directions as “horizontal,” “vertical,” “front,” “rear,” “left,” “right,” and so on are generally made with reference to a particular figure or example and are not necessarily indicative of an absolute orientation or direction. However, relative directional terms for a particular embodiment may generally apply to alternative orientations of that embodiment. For example, “front” and “rear” directions or features (or “right” and “left” directions or features, and so on) may be generally understood to indicate relatively opposite directions or features.

[0023] The following discussion is presented to enable a person skilled in the art to make and use embodiments of the invention. Various modifications to the illustrated embodiments will be readily apparent to those skilled in the art, and the generic principles herein can be applied to other embodiments and applications without departing from embodiments of the invention. Thus, embodiments of the invention are not intended to be limited to embodiments shown, but are to be accorded the widest scope consistent with the principles and features disclosed herein. The following detailed description is to be read with reference to the figures, in which like elements in different figures have like reference numerals. The figures, which are not necessarily to scale, depict selected embodiments and are not intended to limit the scope of embodiments of the invention. Skilled artisans will recognize the examples provided herein have many useful alternatives and fall within the scope of embodiments of the invention.

[0024] As noted above, in some contexts, it may be useful to secure electrical boxes (or other components) at different depths relative to a support structure. For example, in order to comply with requirements that a forward opening of an electrical box be flush with an interior surface of drywall or a ceiling tile, the electrical box may need to be secured to a support structure (e.g., a stud, support bracket, ceiling structure, etc.) at a depth that depends on the depth of the electrical box. Because electrical boxes can be provided with a variety of depths, it may accordingly be necessary to secure different electrical boxes to a particular support structure at different depths relative to the support structure. Embodiments of the invention can provide a bracket and

associated assembly or method that can allow easy adjustment of the mounting depth of an electrical box relative to a support structure, including after the bracket has been installed on the support structure.

[0025] Embodiments of the invention are presented below in the context of particular support structures, including T-grid ceiling support bars and associated brackets. Although these configurations can be particularly useful in some contexts, including due to the particular requirements for mounting electrical boxes to ceiling structures, other configurations are possible. For example, the principles disclosed herein—and embodiments of the invention—can be used with ceiling support structures other than those expressly illustrated or discussed, with non-ceiling support structures (e.g., telescoping or other brackets attached to wall studs), or in a variety of other contexts. Similarly, although particular configurations of electrical boxes are illustrated and discussed below, some embodiments of the invention can be used with electrical boxes having different configurations. And some embodiments of the invention can be used to adjust the mounting depth of components other than electrical boxes.

[0026] Generally, embodiments of the invention can include a bracket that includes a support body and one or more support arms that are collectively configured to support an electrical box relative to one or more other structures (e.g., a ceiling or between-stud support). In some cases, the support arm(s) can be integrally formed with the support body, although other configurations are possible.

[0027] Each support arm can be configured to adjustably receive a fastener, with the fastener extending from the support arm to engage the support body or the electrical box. The fastener can then be adjusted relative to the support arm in order to manually adjust the mounting depth of the electrical box relative to the support body and any associated support structure (e.g., ceiling or between-stud support). Generally, in this regard, adjustment of the fastener can move the electrical box in an axial direction defined by the fastener, relative to at least one of the support body or the associated support arm.

[0028] In some embodiments, the fastener can be a ribbed fastener, such as a fastener with threads or circumferential ridges. For example, some embodiments can include a threaded fastener that is rotatable but not translatable relative to the associated support arm. The threaded fastener can accordingly be rotated by a user in order to cause a translational adjustment of the electrical box. For example, rotation of a threaded fastener may cause translational movement of an electrical box via engagement with a threaded fastener opening on the electrical box or on the support body.

[0029] In some embodiments, it may be useful to arrange the support body and the one or more support arms so that one or more associated fasteners extend within the interior of the electrical box. In some such embodiments, the fasteners may be adjustable from within the electrical box. For example, a support arm may rotatably and non-translatably secure a threaded fastener that extends from the support arm through a threaded fastener opening on the electrical box into the interior of the electrical box. A user can then engage a head of the fastener from within the electrical box, thereby rotating the fastener to cause the electrical box to be translated relative to the support arm via the threaded engagement of the fastener with the fastener opening and the non-translatable engagement of the fastener with the support arm.

[0030] The ability to adjust the mounting depth of an electrical box by engaging a fastener within the interior of the electrical box may be useful, for example, in order to facilitate easier adjustment after the electrical box has been installed (e.g., behind drywall or ceiling tiles). However, in some embodiments, the mounting depth of an electrical box can be adjusted outside of the electrical box. For example, a support arm may be secured to an electrical box and a threaded fastener may extend along the exterior of the electrical box between a rotatable but non-translatable engagement with the support arm and a rotatable engagement with a threaded fastener opening in a support body. A user may then adjust the mounting depth of the electrical box via rotational adjustment of the fastener.

[0031] Referring to FIGS. 1 and 2, an example bracket **100** according to an embodiment of the

invention is installed to adjustably support an electrical box **102** relative to a ceiling structure **104**. In the illustrated example, the ceiling structure **104** includes standard T-grid supports and a set of ceiling brackets **106** extending therebetween. Correspondingly, a support body **108** of the bracket **100** is configured to engage the ceiling brackets **106** and, in particular, is configured to be slidably adjustable relative to the ceiling brackets **106**. In other embodiments, however, other configurations are possible, including configurations other than for ceiling support structures.

[0032] As illustrated in FIG. 2 in particular, the support body **108** includes a central opening **110** that is configured to receive and surround the electrical box **102**. This can provide for easy installation of the electrical box **102** and relatively stable support thereof. In other embodiments, however, other configurations are possible, including configurations in which a support body does not fully surround exterior sides of an electrical box.

[0033] To allow for adjustment of the mounting depth of the electrical box relative to the support body **108** and the ceiling brackets **106**, two support arms **112** extend from opposing sides of the support body **108** to support respective fasteners **114**. In the illustrated embodiment, the support arms **112** are integrally formed with the support body **108** and extend from adjacent to the central opening **110**, in particular from a connection to the support body **108** at an edge of the central opening **110**. This may allow the bracket **100** to be manufactured with particular efficiency, although other configurations are possible.

[0034] In the illustrated embodiment, the extension portions **116** are configured to contact the electrical box **102** between the support body **108** and the support portion **118**, as may help to stabilize the electrical box **102** and guide smooth movement of the electrical box **102** during depth adjustments (e.g., as further described below). In other embodiments, however, other configurations are possible.

[0035] Generally, as discussed above, support arms can support associated fasteners so that adjustment of the fasteners relative to the support arms can adjust the mounting depth of an associated electrical box. In different embodiments, different configurations are possible for support arms and associated fasteners. As illustrated in FIG. 2, for example, each of the support arms **112** includes an extension portion **116** that extends upwardly (or rearwardly, depending on the installation orientation) along an exterior side of the electrical box **102**. Each of the support arms **112** also includes a support portion **118** that angles (e.g., at 90°) away from the extension portion **116** at a bend opposite the support body **108**.

[0036] Thus arranged, the support portion **118** can support the associated fastener **114** exterior to and in axial alignment with (i.e., relative to the axial direction of the fastener **114**) the rear wall of the electrical box **102**. In particular, the support portions **118** extend behind the rear wall of the electrical box **102** at opposing corners thereof (i.e., at corners not included on a common edge of the rear wall) so that the fasteners **114** extend through opposing corners of the rear wall of the electrical box **102**. As also generally discussed below, this may help to provide even, easy, and reliable adjustment of the depth of the electrical box **102** while also helping to prevent the fasteners **114** from unduly interfering with wires or electrical devices (not shown) within the electrical box **102**.

[0037] Continuing, the fasteners **114** extend from the support portions **118** of the support arms **112** through fastener openings **120** in the rear wall of the electrical box **102**. Thus, via rotation of the fasteners **114**, the electrical box **102** can be moved in the axial direction of the fasteners **114** in order to adjust the mounting depth of the electrical box **102** relative to the support body **108**. In particular, in the illustrated embodiment, the fasteners **114** are rotatable relative to the support portions **118** but are translationally (axially) fixed relative to the support portions **118** (e.g., via riveting or peening of the fasteners, use of E-, C-, or other clips, use of integral or removable collars, or other techniques). The fasteners **114** are also threadedly engaged with threads on extruded features at the fastener openings **120**. Accordingly, when the fasteners **114** are rotated, the threaded engagement of the fasteners **114** with the fastener openings **120**, and the rotatable but non-

translational engagement of the fasteners **114** with the support portions **118**, can collectively cause the electrical box **102** to move axially along the fasteners **114**. Thus, via rotation of the fasteners **114** and corresponding axial movement of the fasteners **114** through the fastener openings **120**, the electrical box **102** can be secured at any of a plurality of depths relative to the support body **108**. Indeed, because of the threaded configuration of the fasteners **114**, the electrical box **102** can be secured at any depth along a continuous range, as defined by the available length of the fasteners **114**.

[0038] As discussed herein, unless otherwise indicated, “translationally fixed” and the like does not necessarily indicate an absolutely fixed arrangement, in which zero translational movement is permitted. Rather, some translationally fixed components (e.g., the fasteners **114**) may be able to move translationally somewhat, but only to a certain amount and generally not so as to provide meaningful translational adjustment of a corresponding component (e.g., an electrical box). For example, depending on how the fasteners **114** are fixed to the support portions **118**, the fasteners **114** may be able to move slightly in the axial direction, relative to the support portions **118** (e.g., by 1-5 mm or 1-3% of the axial length of the fasteners **114** or of the total depth of the electrical box **102**).

[0039] In the illustrated embodiment, heads of the fasteners **114** are disposed entirely within the interior of the electrical box **102**, via the extension of the fasteners **114** from the support portions **118** through the fastener openings **120**. This arrangement may be useful in some cases, because the open front of the electrical box **102** may remain readily accessible even after other surrounding structures (e.g., ceiling tiles or drywall) have been installed around the electrical box **102**. Accordingly, for example, even after ceiling tiles have been installed for the ceiling structure **104** of FIGS. **1** and **2**, a user may still easily manually adjust the depth of the electrical box **102** by accessing the fasteners **114** via the front opening of the electrical box **102**. Further, disposing the fasteners **114** behind and within, rather than alongside, the electrical box **102** can help to reduce the overall footprint of the bracket **100** and electrical box **102** as installed (i.e., relative to an area in a plane perpendicular to the axis of the fasteners **114**) and avoid interference between the fasteners are external features (e.g., features of a building support structure). In other embodiments, however, including as discussed below, other configurations are possible.

[0040] In some embodiments, other aspects of the configuration illustrated in FIGS. **1** and **2** can also be changed while preserving similarly beneficial adjustability. For example, in some embodiments, fasteners may be translationally fixed relative to a support arm and may engage a threaded feature at a different location on or in an electrical box. Or a fastener threadedly engaged with a support arm and translationally fixed but rotatable relative to an electrical box. Similarly, although configurations with a set of opposing support arms (e.g., at opposing corners) may sometimes provide a particularly optimal balance between stability and ease of adjustment, some configurations may have a different number of support arms, or one or more support arms disposed differently relative to a support body than is illustrated in FIGS. **1** and **2**.

[0041] As another example, FIG. **3** illustrates a bracket **150** that is configured similarly to the bracket **100**, so that the mounting depth of an electrical box **152** relative to a ceiling structure **154** can be easily manually adjusted from the interior of the electrical box **152**. In some aspects, however, the bracket **150** differs from the bracket **100**. For example, a support body **158** of the bracket is configured to engage telescoping ceiling brackets **156**, of a different configuration than the ceiling brackets **106**.

[0042] In some embodiments, a support arm may be separate from a support body, so that adjustment of a fastener moves the support arm relative to the support body and thereby adjusts a mounting depth of an associated mounting bracket. A bracket **200** of this type according to an embodiment of the invention is in FIGS. **4** and **5**.

[0043] Similarly to the brackets **100**, **150**, the bracket **200** includes a support body **208** that is configured to surround an electrical box **202** and support the electrical box **202** relative to a support

structure (e.g., ceiling brackets **206**). In contrast to the brackets **100**, **150**, however, support arms **212** of the bracket **200** are separated and spaced apart from the support body **208** and are secured to exterior sides of the electrical box **202** (e.g., directly attached to the box **202**, using welding or fasteners, or integrally formed with the box **202**).

[0044] Similarly to the support arms **112**, the support arms **212** are configured to retain threaded fasteners **214** so that the fasteners **214** are translationally fixed but rotatable relative to the support arms **212**. In contrast to the fasteners **114**, however, the fasteners **214** extend from the support arms **212** to engage threaded fastener openings **220** on the support body **208** rather than on the electrical box **202**. Accordingly, as the fasteners **214** are rotated relative to the support arms **212**, the fixed engagement of the support arms **212** with the electrical box, the non-translational engagement of the fasteners **214** with the support arms **212**, and the threaded engagement of the fasteners **214** with the support body **208** cause the electrical box **202** to translate relative to the support body **208** (i.e., in the axial direction of the fasteners **214**). Thus, via rotation of the fasteners **214** and axial movement of the fasteners **214** through the fastener openings **220**, the electrical box **202** can be secured at any of a plurality of depths relative to the support body **208**.

[0045] As also noted above, in different embodiments, different arrangements can be used to secure fasteners to support arms. In the illustrated embodiment, for example, each of the support arms **212** includes a stamped fold **222** that forms a clip to fix a head of the associated fastener **214** against translation while still allowing the fastener **214** to rotate. Further, an access opening **224** is provided on a forward side of each of the folds **222** (relative to the orientation of the electrical box **202**) so that a hand tool can be used to manually adjust each of the fasteners **214** without removing the fastener **214** from the fold **222**.

[0046] Also in contrast to the configurations of the brackets **100**, **150**, the fasteners **214** of the bracket **200** extend along exterior sides of the electrical box **202**. In some embodiments, this arrangement may also provide notable benefits. For example, because the fasteners **214** do not extend within the electrical box **202**, the entire interior volume of the electrical box **202** remains available to receive and arrange other components.

[0047] In some embodiments, as also discussed above, parts of a bracket can be configured to extend along (e.g., contact) one or more sides of an electrical box in order to help to stabilize the electrical box or to help guide depth adjustments thereof. Accordingly, for example, some embodiments similar to the bracket **200** can include structures similar to the extension portions **116** of the bracket **100** (see, e.g., FIG. 2), or other features, that extend from adjacent to a central opening of a bracket (or elsewhere) in order to contact and support an electrical box secured to the bracket.

[0048] Although the use of multiple adjustable fasteners can provide some benefits relative to stability, some embodiments may use only a single adjustable fastener. For example, as illustrated in FIG. 6, a bracket **250** is configured to adjustably support an electrical box **252** relative to ceiling brackets **256**. The bracket **250** is generally similar to the bracket **100** (see, e.g., FIG. 2), with a support body **258** and two extension portions **266** that extend on opposing sides of the electrical box **252**. In contrast to the bracket **100**, however, the bracket **250** includes only a single support portion **268** that extends between the extension portions **266** across the back wall of the electrical box **252** (i.e., the extension portions do not form independent support arms). Correspondingly, only a single fastener **264**, riveted to the support portion **268**, extends through a central threaded fastener opening **270** in the electrical box **252**, so that a mounting depth of the electrical box **252** can be adjusted by rotating the single fastener **264**. In other embodiments, multiple fasteners can be used with a similar bracket, or a single fastener can be used with brackets that are configured differently (e.g., do not have a bridging support portion as shown for the support portion **268**).

[0049] FIGS. 7 and 8 illustrate a bracket **300** that is generally similar in design and functionality to the brackets **100**, **150** of FIGS. 1 and 3. Like the brackets **100**, **150**, the bracket **300** is configured so that the mounting depth of an electrical box **302** relative to a support structure (e.g., telescoping

ceiling brackets **306** of FIG. **8**) can be easily manually adjusted from the interior of the electrical box **302** and discussion of brackets **100**, **150** above generally also applies to the bracket **300** (and vice versa). In some aspects, however, the bracket **300** differs from the brackets **100**, **150**. For example, as also discussed below, a support body **308** of the bracket **300** includes bendable tabs **326** (only one shown) that are configured to engage the ceiling brackets **306** (see, e.g., FIG. **8**). Further, the support body **308** also includes sets of attachment arms, here configured as exterior arms **332** of the support body **308** that support the bendable tabs **326** and as resilient interior spring arms **330**, disposed on opposite sides of each of the exterior arms **332**.

[0050] As best seen in FIG. **8**, the bracket **300** is configured to be mounted to the ceiling brackets **306** with the support body **308** contacting the ceiling brackets **306** on a first side **334**, so that the ceiling brackets **306** are gripped between sets of the arms **330**, **332** on each of two opposing sides of the bracket **300**. With the support body **308** thus disposed, the bendable tabs **326** can then be bent from an installation configuration (see FIG. **7**) toward the electrical box **302** (and the corresponding spring arms **330**) and into an installed configuration, in which the tabs **326** overlay an the ceiling brackets **306** on an opposing second side **336** and thereby further secure the support body **308** to the ceiling brackets **306**. As a result, the support body **308** of the bracket **300** can be easily secured to the ceiling brackets **306** without separate fasteners and, in particular, is configured to be slidably adjustable relative to the ceiling brackets **306** once so secured. Of note, although the bracket **300** can be secured to a support structure without using fasteners, including as discussed above, it may nonetheless be useful to further secure the bracket **300** with threaded fasteners (e.g., as inserted through a front face of the attachment arm **332** or the support body **308**).

[0051] In the embodiment illustrated, the first side **334** of the ceiling bracket **306** is a lower (bottom) side thereof, and the second side **336** is an upper (top) side thereof. Thus, the bracket **300** can be mounted to ceiling brackets **306** from below, which may provide an improved installation process relative to conventional designs that must be installed from above. However, other installation orientations are possible, including installations in which the support body **308** is oriented vertically (e.g., as attached to a between-stud telescoping bracket (not shown)) or installed from above a support structure (e.g., similarly to the brackets **100**, **150**, **200**, **250**). Further, although the bracket **300** of the FIGS. **7** and **8** includes two bendable tabs **326** asymmetrically disposed on the exterior attachment arm **332**, other numbers or configurations of bendable tabs may be used in other embodiments.

[0052] FIGS. **9** through **11** illustrate another bracket **400** configured to allow manual adjustment of a mounting depth of an electrical box **402** relative to a support structure (e.g., telescoping between-stud brackets **406** of FIG. **11**), from the interior of the electrical box **402**. The bracket **400** is generally similar in design and functionality to the bracket **300** of FIGS. **7** and **8**, and discussion of bracket **300** above generally also applies to the bracket **400** (and vice versa). In this regard, for example, a support body **408** includes a body opening **410** that is fully enclosed by surrounding material of the support body **408** (see FIG. **10**), such as may provide for improved strength and durability of the bracket **400**. Further, two independent support arms **412** extend integrally from opposing edges of the body opening **410**, each with an extension portion **414** and a distal bend that leads to a free end proximate a support portion **416**.

[0053] Continuing, threaded fasteners **418** are riveted to the support portions **416** to be rotatably supported in alignment with opposing rear corners of the electrical box **402**, with heads of the fasteners **418** disposed inside of the electrical box **402** for adjustment (see FIG. **10**). As also shown in FIG. **10** in particular, the fasteners **418** are thus generally constrained to move only axially (and rotationally about their own axes) relative to the interior of the electrical box **402**, and to move axially only along a corresponding translation zone **420** of relatively limited size (e.g., extending only 25% of the width of the fasteners **418** beyond the fasteners **418** in any direction perpendicular to the rotational axes of the fasteners **418**). Thus, as also discussed above, the fasteners **418** may impose relatively minimal reductions in available space for electrical devices within the electrical

box **402**. Further, the translation zones **420** being of relatively small area and being disposed on opposing corners of the electrical box **402** can allow for easy adjustment of the fasteners **418** despite the presence of electrical-device mounting tabs **404** on other corners of the electrical box **402** (or elsewhere on other boxes).

[0054] As a further example, referring again to FIG. **9**, the support arms **412** taper from a perspective moving from the body opening **410** toward the support portions **416**. As also discussed below, this taper (and others) can allow for more economical manufacturing of the disclosed brackets, while also ensuring appropriate overall strength for the support arms. Additionally, tapered support arms can improve accessibility an electrical boxes supported by the brackets. For example, as illustrated for the bracket **400**, the taper of the support arms **412**—as shown, from more than half to less than a quarter of the widths of the body opening **410** and the electrical box **402**—can allow users to easily access all four large rear knock-outs **422** of the electrical box **402** and two of the large knock-outs **424** on the corresponding lateral side of the electrical box **402**. Thus, despite providing relatively secure and highly adjustable support, the support arms **412** can still allow users to configure the electrical box **402** with a high degree of flexibility, including relative to use of the knock-outs **422**, **424**.

[0055] As still another example, the support arms **412** can also provide a far-side support for the bracket **400** or for a larger bracket assembly. As illustrated in FIG. **11**, for example, the support arms **412** extend rearward from the between-stud bracket **406** so that the support portions **416** thereof are adjacent to a far-side (e.g., drywall) surface. Thus, the support arms **412** can help to prevent the brackets **400**, **406** or the electrical box **402** from being pushed rearwardly relative to the larger support structure. In this regard, lips **426** along the outer edges of the support can also be useful: not only can the lips **426** provide improved structural rigidity for the support arms **412**, but they can also create a recessed area **428** (see FIG. **9**) within which the riveted end of the fasteners **418** can freely rotate without interference from any far-side structure. Further in this regard, the independent, spaced-apart, opposite-corner configuration of the support portions **416** can contribute to particularly stable far-side support functionality (e.g., in contrast to a single, bridging support member that extends between mid-points of opposing lateral sides of an electrical box).

[0056] Despite many similarities, the bracket **400** also differs from the bracket **300** of FIGS. **7** and **8** in some aspects. For example, to further guide movement of the electrical box **402** during depth adjustment, the support body **408** also includes guide tabs **430**. As illustrated in FIG. **9** in particular, the guide tabs **430** extend integrally from the body, adjacent to attachment arms **432**, **434** (with bendable tabs **436**), into alignment with the body opening **410** (i.e., to overlap with a rearward projection of the area of the body opening **410**). Accordingly, when the electrical box **402** is slidably disposed within the body opening **410**, the guide tabs **430** resiliently engage the corresponding side walls of the electrical box **402** and thereby assist to maintain an aligned and smooth movement of the electrical box **402** within the body opening **410** during depth adjustment. Thus, in the illustrated embodiment, the electrical box **402** can be laterally stabilized on all four sides, by a combination of the support arms **412** and the guide tabs **430**, so that relatively smooth adjustment and stable anchoring of the electrical box **402** can be achieved.

[0057] In some embodiments, brackets according to the invention can be formed from stamping of single-piece blanks. As opposed to some conventional approaches, this may improve manufacturing efficiency, reduce manufacturing costs, and provide brackets of higher and more reliable quality than other methods. Further, integrally-formed brackets can be easier to manage during installation and adjustment, and can allow for improved inventory efficiency.

[0058] In this regard, for example, FIG. **12** illustrates a manufacturing blank **500** that can be formed, through a series of stamping operations, into the bracket **400**. In particular, the blank **500** includes a single-piece planar body **502** that may be initially stamped from a larger piece of sheet metal and that fully encloses a central body opening **504**. The body opening **504** is generally rectangular (e.g., square), with a rectangular perimeter profile that is interrupted along opposing

edges 506 by opposing supports 508 that extend from the edges 506 into the body opening 504, although other opening shapes are also possible. Further, the supports 508 taper, from a perspective moving from the edges 506 towards the distal ends of the supports 508, from a maximum width 510 that is more than 50% of a corresponding width of the body opening 504 (e.g., 67%) to a width 512 of less than 25% of the width of the body opening 504 (e.g., 22%) at a location corresponding to openings 524 that will support a threaded fastener for adjustment of the electrical box. Overall, the supports 508 thus occupy most (e.g., 75%, 80%, or more) of the area of the body opening 504. In some cases, and in other similar configurations, this can provide substantial material efficiencies for manufacturing. As also discussed above, it can also allow the resulting support arms 412 to provide access to the knock-outs 422, 424 (see, e.g., FIG. 9) and help to ensure appropriate overall strength when the supports 508 are formed into the support arms 412 (see, e.g., FIG. 9), particularly when edge regions 508a of the supports 508 are folded to form the lips 426 (see, e.g., FIG. 9).

[0059] The body 502 also includes sets of arms 514 on opposing sides thereof, extending from an outer edge of a main part of the body 502, in a direction away from the body opening 504. In the illustrated embodiment, the arms 514 extend from different opposing sides of the body 502 and in different opposing directions than do the supports 508, as may allow for particularly effective collective engagement of an electrical box and a support structure by features formed by the arms 514 and the supports 508. In other embodiments, however, other configurations are possible.

[0060] Although a variety of arm configurations are possible, in the illustrated embodiment the sets of arms 514 are symmetrical relative to each other, about a central axis of the blank 500, and each set of arms 514 is internally symmetrical about a perpendicular axis, except for tab extensions 516 that correspond to the bendable tabs 436 (as further discussed below), such as may result in the bracket 400 exhibiting particularly reliable stability. In particular, each of the sets of arms 514 includes a central attachment arm 518 that extends farther from the main portion of the body 502 (and the body opening 504) than opposing peripheral attachment arms 520, with the tab extension 516 extending outwardly from one lateral side of the central attachment arm 518. Additionally, guide arms 522 extend by the same distance as the central attachment arm 518, between the central attachment arm 518 and the peripheral attachment arms 520.

[0061] As further detailed below, the attachment arms 520 correspond to the attachment arms 432 (see, e.g., FIG. 9), the attachment arms 518 correspond to the attachment arms 434, and the guide arms 522 correspond to the guide tabs 430. Thus, with continue reference also to FIG. 9, the illustrated spatial arrangement of the sets of arms 514 may be particularly beneficial, including so that there is an appropriately wide spacing between the attachment arms 432, so that the attachment arms 434 are sufficiently long to extend around a support structure when in a bent configuration, and so that the guide tabs 430 are sufficiently long to extend into the body opening 410 to engage the electrical box 402. However, a variety of other arm configurations are also possible.

[0062] In some cases, other beneficial features can also be provided. For example, anchor extensions 526 on the attachment arms 518 can provide attachment locations for cables or other devices to further help support the bracket 400 relative to support structures. As another example, alignment features such as indexing indents 528 can help to appropriately align a centerline of an electrical box during installation. Mounting openings 530 can also help with visual alignment or can serve as anchor points for fasteners that are used to further secure the bracket 400 in place.

[0063] In some implementations, devices or systems disclosed herein can be utilized, manufactured, or installed using methods embodying aspects of the invention. Correspondingly, any description herein of particular features, capabilities, or intended purposes of a device or system is generally intended to include disclosure of a method of using such devices for the intended purposes, of a method of otherwise implementing such capabilities, of a method of manufacturing relevant components of such a device or system (or the device or system as a whole), and of a method of installing disclosed (or otherwise known) components to support such

purposes or capabilities. Similarly, unless otherwise indicated or limited, discussion herein of any method of manufacturing or using for a particular device or system, including installing the device or system, is intended to inherently include disclosure, as embodiments of the invention, of the utilized features and implemented capabilities of such device or system.

[0064] In this regard, some embodiments can include a method of manufacturing for a bracket (e.g., an integrally formed bracket) for adjustable-depth mounting of electrical boxes. For example, still referring to FIGS. 9 and 12, a method of manufacturing the bracket 400 can include initially stamping the shape of the blank 500 from a single piece of metal material. Further operations can then be executed on the blank 500 to form the edge regions 508a of the supports 508 into the lips 426, to bend the supports 508 into the support arms 412. The sets of arms 514 can also be bent, including multiple bends, respectively, to form the attachment arms 518, 520 and the guide arms 522, into the attachment arms 432, 434 and the guide tabs 430. Generally, the tab extension 516 can be left as the guide tab 430 in an un-bent configuration until installation, when the guide tab 430 can be used to secure the bracket 400 as needed (e.g., as generally discussed above). The various bends noted above can be executed in different orders, in different implementations, including with one or more bends executed simultaneously (i.e., as part of a single stamping step).

[0065] The discussion above is framed relative to particular brackets and associated arrangements. However, those of skill in the art will recognize that this discussion implicitly also discloses various methods of adjustably mounting electrical boxes relative to support structures. Similarly, as also discussed above, the particular configurations of the support bodies and other components expressly described and illustrated in the various embodiments are presented as examples only, and the concepts disclosed herein can be used to adjustably secure electrical boxes (or other components) relative to a variety of bracket configurations and support structures. In this regard, for example, brackets according to some embodiments can be configured for attachment to between-stud supports, as integral parts of between-stud supports, for attachment directly to building studs or other building structures, and so on. Further, specific features discussed in detail relative to certain embodiments can be generally configured or used similarly with other embodiments, including relative to similar features on those embodiments or as substitutions or additions to those embodiments.

[0066] Thus, embodiments of the invention can provide improved brackets for adjustable mounting of electrical boxes. For example, some embodiments can provide brackets that allow an electrical bracket to be adjusted among any number of different mounting depths relative to a support structure, including any number of different mounting depths along a continuous range of depths.

[0067] The previous description of the disclosed embodiments is provided to enable any person skilled in the art to make or use the invention. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments without departing from the spirit or scope of the invention. Thus, the invention is not intended to be limited to the embodiments shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. A bracket assembly for adjustable mounting of an electrical box, the bracket assembly comprising: a support body including a body opening to receive the electrical box; a first support arm extending from the support body, the first support arm including a first support portion for a first fastener to engage the electrical box to adjust a position of the electrical box with respect to the support body, the first support arm extending from a first side of the body opening; a second support arm extending from the support body, the second support arm including a second support portion for a second fastener to engage the electrical box to adjust the position of the electrical box with respect to the support body, the second support arm extending from a second side of the body

opening, opposite the first side; and a center axis extending through the first and second side of the body opening; wherein the first support arm and the second support arm extend asymmetrically about the center axis.

2. The bracket assembly of claim 1, wherein the first support portion extends from the first support arm behind a rear wall of the electrical box; and wherein the second support portion extends transversely from the second support arm behind the rear wall of the electrical box.

3. The bracket assembly of claim 1, wherein the first fastener is axially fixed relative to the first support portion and is rotatable relative to the first support portion; and wherein the second fastener is axially fixed relative to the second support portion and is rotatable relative to the second support portion.

4. The bracket assembly of claim 1, wherein a rear wall of the electrical box includes fastener openings; and wherein the electrical box is axially movable relative to the first fastener and the second fastener at the fastener openings.

5. The bracket assembly of claim 1, wherein the first and second fastener are adjustable from inside the electrical box.

6. The bracket assembly of claim 1, wherein the second support portion is arranged diagonally opposite the first support portion relative to the body opening.

7. The bracket assembly of claim 1, further comprising a plurality of guide tabs that extend integrally from the support body to biasingly engage side walls of the electrical box.

8. The bracket assembly of claim 7, wherein one or more of the guide tabs extend from a third side of the support body.

9. The bracket assembly of claim 1, wherein the first support arm tapers, such that a width of the first support arm decreases between the support body and the first support portion.

10. A bracket assembly for adjustable mounting of an electrical box relative to a support structure, the electrical box including a rear wall, the bracket assembly comprising: a support body to be secured to the support structure, the support body including a body opening to receive the electrical box; and a first support arm and a second support arm extending from the support body on opposing sides of the body opening, the first support arm extending independently relative to the second support arm, the first support arm having a first support portion extending partly across a rear wall of the electrical box, the first support portion receiving a first fastener that engages the electrical box, and the second support arm having a second support portion extending partly across a rear wall of the electrical box.

11. The bracket assembly of claim 10, wherein the first fastener is axially fixed to the first support arm and rotatable relative to the first support arm, the first fastener extending through and being axially movable relative to a first fastener opening in the rear wall of the electrical box.

12. The bracket assembly of claim 10, wherein the first fastener moves axially relative to the first fastener opening upon rotation of the first fastener relative to the first support arm.

13. The bracket assembly of claim 10, wherein the first support arm extends from a first side of the body opening, wherein the second support arm extends from a second side of the body opening, and wherein the first and second support arms extend asymmetrically from the support body about a center axis that extends through the first and second side of the body opening.

14. The bracket assembly of claim 10, wherein the first support arm tapers, such that a width of the first support arm decreases between the support body and the first support portion.

15. The bracket assembly of claim 10, wherein the first support portion extends transversely relative to the first support arm.

16. The bracket assembly of claim 10, and further comprising: a first guide tab; and a second guide tab; wherein the first guide tab extends to resiliently engage one of a plurality of side walls of the electrical box, and wherein the second guide tab extends to resiliently engage one of the plurality of side walls of the electrical box.

17. The bracket assembly of claim 16, wherein the first support arm extends from a first side of the

body opening, wherein the second support arm extends from a second side of the body opening, and wherein the first guide tab is located on a third side of body opening.

18. A method of producing a bracket assembly for adjustable mounting of an electrical box relative to a support structure, the method comprising: stamping a blank from metal into a shape, the blank including a support body with a body opening and a first support arm and a second support arm extending from the support body on opposite sides of the body opening; bending the first support arm and the second support arm at the body opening to extend the first support arm and the second support arm away from the support body; forming a first bend in the first support arm to form a first support portion to extend partly across the body opening; forming a second bend in the second support arm to form a second support portion to extend partly across the body opening; inserting a first fastener in the first support portion to be engageable with the electrical box to move the electrical box with respect to the support body; and inserting a second fastener in the second support portion to be engageable with the electrical box to move the electrical box with respect to the support body.

19. The method of claim 18, wherein the first support arm and the second support arm are bent at locations that are disposed diagonally opposite each other relative to the body opening.

20. The method of claim 18, and further comprising stamping the blank from the metal to form a first guide tab; and bending the first guide tab to extend past a first edge of the body opening to be resiliently engageable with a first side wall of the electrical box.
